Dr Caitlin D. Kuempel

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Career Highlights

I have an h-index of 21, with 43 peer-reviewed papers (17 as lead author). Twelve are in the top 10% most cited publications worldwide, including 28 publications in the top 10% journals (e.g., Nature Sustainability, PNAS, Current Biology). Over 85% of my publications involve international collaboration. In the past 2 years, I have attracted project funding of >$1.2 million. I collaborate frequently with industry on applied conservation challenges; I was the Monitoring and Data Management lead for the World Wildlife Fund's Coral Reef Rescue Initiative and currently lead the cumulative impacts workstream of the Blue Economy Cooperative Research Centre Marine Spatial Planning project. I teach into five courses at Griffith, and convene two (~140 students), with consistently high student feedback scores (>4 out of 5). I am passionate about mentoring and science communication. I currently supervise a post-doctoral fellow and three HDR candidates. I am co-organising Brisbane's Soapbox Science event for National Science Weeek and sit on the steering committee for the International Congress of Conservation Biology to be held in Brisbane in 2025. I am well-known for teaching R programming (>400 trained) and Marxan decision support tool (>200 trained). I have >200,000 reads in The Conversation, have been featured in general media such as live interviews on Triple J's Hack program and interviews in The Guardian, and was a semi-finalist in the FameLab science communication competition. I serve as an Early Career Representative for the School of Environment and Science and participated in Griffith's Women in Leadership program.

Education

PhD Conservation Science, The University of Queensland, St. Lucia, QLD, Australia — Sept. 2018

Thesis: Evaluating trade-offs between quantitative and qualitative protected area objectives
Supervisors: Prof. Hugh Possingham, Prof. Catherine Lovelock and Dr. Alienor Chauvenet
Dean's Award for Outstanding HDR thesis (ranked within top 10% in field by examiners).

M.S. Marine Biology Northeastern University, Nahant, MA — Dec. 2013

Coral Reef Ecology (GPA 4.0/4.0), Supervisor: Dr Andrew ALtieri

BSc Environmental Science (Honours), BA French, University of Iowa, Iowa City, IA — May 2012

Honors in coral morphometrics, Certificate in Sustainability

Professional Experience

Lecturer, Marine Ecosystem Modelling, Griffith University, Brisbane, Australia — June 2022-present

Balanced profile: 40% teaching, 40% research, 20% service

Post-doctoral Fellow, ARC CoE for Coral Reef Studies, UQ, Brisbane, Australia — Feb. 2020-May 2022

Explored ways to manage and monitor climate resilient reefs to benefit people and nature as part of the World Wildlife Fund's Coral Reef Rescue Initiative

Post-doctoral Fellow, NCEAS, Santa Barbara, CA — Sept. 2018-Dec. 2019

Quantified, compared and mapped the production and environmental impacts of all major food systems as part of a collaborative working group

Conservation Planning Consultant, IUCN and The University of Queensland — July-Aug.2018

Collated data and developed conservation plans for four Pacific Island nations using the spatial conservation planning software, Marxan.

Research Assistant: Smithsonian Tropical Research Institute, Bocas del Toro, Panama April-Oct. 2014

Thalassia testdinum Experimental Network (TEN): investigated the impacts of nutrients and grazer abundance on seagrass across a latitudinal gradient (Florida, Belize, Panama).

Conservation Intern: Central Caribbean Marine Institute, Little Cayman, Cayman Islands Jan.-April 2014

Assisted with lionfish management research, coral nursery, and climate data collection. Provided weekly educational talks and tours to guests and tourists.

Short-Term Fellow: Smithsonian Tropical Research Institute, Bocas del Toro, Panama June-Dec. 2013

Investigated the role of consumers in coral-algal community dynamics under the supervision of Dr. Andrew Altieri.

Research Assistant: University of Iowa Department of Geoscience, Iowa City, IA May 2009-2010

Produced thin-sections of coral specimens and performed morphemetric analysis of the coral species Diploria, resulting in a final honors thesis and research presentation under the supervision of Dr. Ann Budd.

Peer-Reviewed Publications

I have 50 peer-reviewed publications (h-index 22), 17 as lead author, >1700 citations. International collaboration on ~85% of publications. See Web of Science, Google scholar or ORCID

44. **Kuempel, C.D.**, Christofidis, M., Fuller, R., Bush, R., Garnett, S.T., Jackson, M.V. (submitted). Half of coastal aquaculture sites in Queensland are in close proximity to shorebird habitats. *Ocean and Coastal Management*
43. Christofidis, M., Lyons, M., **Kuempel, C.D.** (2025). One-third of Australia's coastal terrestrial aquaculture at risk from sea level rise. *Aquaculture Science and Management*
42. Stockbridge, J.S., Brown, C.J., **Kuempel, C.D.** (under revision). Mapping the global co-location potential of offshore wind energy and aquaculture production. *Ocean and Coastal Management* (IF = 4.3)
41. Vozzo, M., Christofidis, M. Griffiths, L., Kelly, R., Manion, M.L., Barmand, S., Bolin, J., Fuenzalida-Miralles, P., Harris, M., Issell, A., *Kuempel, C.D.*, Diaz, M., Murunga, M., Palmer, S., Schaefer, N., Simpson, J. (2025). Advancing ocean sustainability through better science integration: perspectives of Early Career Ocean Professionals *Front. in Ocean Sust.*
40. **Kuempel, C.D.**, Stockbridge, J.S., O'Neill, T., Griffiths, L., Frid,C.L.J., Brown, C.J. (2025). Frameworks and best practices for cumulative impact accounting in offshore wind energy development. *Marine Policy* (IF = 3.8)
39. Fleury, A., O'Hara, C.C., Butt, N., Restrepo, J., Halpern, B.S., Klein, C.J., **Kuempel, C.D.**, Gaynor, K.M., Bentley, L.K., Richardson, A.J., Dunn, D.C. (2024). Spatial and life history variation in a trait-based species vulnerability and impact model. *PlosOne* (IF = 3.8)
38. **Kuempel, C.D.**, Arnett, E., Klein, C.J. (2024). Quantifying global redundant fisheries trade to streamline seafood supply chains. *PlosOne* (IF = 3.8)
37. **Kuempel, C.D.** and Suarez-Castro, A.F. (2024) The Global Biodiversity Framework can be leveraged to better manage transboundary watersheds. *Proceedings of the National Academy of Sciences* (IF = 12)
36. **Kuempel, C.D.**, Thomas, J., Wenger, A., Jupiter, S.D., Suarez-Castro, A.F., Nasim, N., Klein, C.J., Hoegh-Guldberg, O. (2024). Assessing opportunities for improved sanitation to support coral reef conservation. *Environmental Pollution* (IF = 8.9)

35. Vella, T., Roberson, L., **Kuempel, C.D.**, Klein, C (2023). Quantifying the accessibility of sustainable seafood in South East Queensland, Australia. *Marine and Freshwater Research* (IF = 2.3)

34. Adams, V., Chauvenet, A.L.M., Stoudmann, N., Gurney, G.G., Brockington, D., **Kuempel, C.D.** (2023). Multiple-use protected areas are critical to equitable and effective conservation. *One Earth* (IF = 16.7)

33. Halpern, B.S., Frazier, M.,Rayner, P.E., Clawson, S.G., Blanchard, J.L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Jacobsen, N.S., **Kuempel, C.D.**, M., Moran, D., Nash, K.L., Williams, D.R. (2023). Matters Arising - Diving deeper into fisheries disturbance. *Nature Sustainability* (IF = 29.2)

32. Froehlich, H., Montgomery, J.C., Williams, D.R., O'Hara, C., **Kuempel, C.D.**, Halpern, B.S. (2023). Biological life history and farming scenarios of marine aquaculture help reduce wild marine fishing pressure. *Fish and Fisheries* (IF = 7.3)

31. Wakwella, A., Wenger, A., Jenkins, A., Lamb, J., **Kuempel, C.D.**, Claar, D., Corbin, C., Falinski, K., Rivera, A., Grantham, H.S., Jupiter, S.D. (2023). Integrated watershed management solutions for healthy coastal ecosystems and people. *Cambridge Prisms: Coastal Futures*

30. **Kuempel, C.D.**, Frazier, M., Verstaen, J., Rayner, P.E., Blanchard, J.L., Bouwman, L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Jacobsen, N.S., McIntyre, P.B., Metian, M., Moran, D., Nash, K.L., Tobben, J., Williams, D.R., Halpern, B.S. (2023). Feed drives land-sea overlap of the global cumulative environmental pressure of farmed chicken and salmon. *Current Biology* (IF = 11)

29. Beger, M., Metaxas, A., Balbar, A.C., McGowan, J.A., Daigle, R., **Kuempel, C.D.**, Treml, E.A., Possingham, H.P. (2022). Demystifying ecological connectivity for actionable spatial conservation planning. *Trends in Ecology and Evolution* (IF = 18.1 citations = 9)

28. Halpern, B.S., Frazier, M., Verstaen, J., Rayner, P.E., Clawson, S.G., Blanchard, J.L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Jacobsen, N.S., **Kuempel, C.D.**, McIntyre, P.B., Metian, M., Moran, D., Nash, K.L., Tobben, J., Williams, D.R. (2022). The cumulative environmental footprint of global food production. *Nature Sustainability* (IF = 29.2, citations 20)

27. Hicks, C.C. Gephart, J.A., Koehn, J.Z., Nakayama, S., Payne, H., Allison, E.H., Delhabib, D., Cao, L., Cohen, P.J., Fanzo, J., Fluet-Chouinard, E., Gelcich, S., Golden, C.D., Gorospe, K.D., Isaacs, M., **Kuempel, C.D.**, Lee, K.N., MacNeil, M.A., Maire, E., Njuki, J., Rao, N., Sumaila, U.R., Selig, E.R., Thilsted, S.H., Wabnitz, C.C.C., Naylor, R.L. (2022). Towards Justice in Blue Food Systems. *Nature Food* (IF = 23.2, citations 14)

26. Klein, C.J., **Kuempel, C.D.**, Watson, R.A., Teneva, L., Coll, M., Mora, C. (2022). Global fishing between jurisdictions with unequal fisheries management. *Environmental Research Letters* (IF = 7.8, citations 5)

25. Almaraz, M., **Kuempel, C.D.**, Salter, A., Houlton, B.Z., Halpern, B.S. (2022). The impact of excessive protein consumption on human wastewater nitrogen loading of US waters. *Frontiers in Ecology and the Environment* (IF = 12.2, citations 5)

24. Clawson, S.G.,**Kuempel, C.D.**, Blasco, G., Froehlich, H.E., Cottrell, R.S., Metian, M., Tobben, J., Williams, D.R., Nash, K.L., Frazier, M., Verstaen, J., Halpern, B.S. (2022). Mapping the spatial distribution of global mariculture production. *Aquaculture* (IF = 4.6, citations 12)

23. Williams, B.A., Watson, J.E.M., Beyer, H.L., Klein, C.J., Montgomery, J., Runting, R.K., Roberson, L.A., Halpern, B.S., Grantham, H.S., **Kuempel, C.D.**, Frazier, M., Venter, O.,

Wenger, A. (2021). The global rarity of intact coastal regions. *Conservation Biology* (IF = 6.5, citations 39)

22. **Kuempel, C.D.**, Simmons, B.A., Davey, M.(2022) Assessing he status of existing and tentative marine natural World Heritage areas reveals opportunities to better achieve World Heritage Convention goals. *Journal of Environmental Management* (IF = 8.4, citations 1)

21. **Kuempel, C.D.**, Tulloch, V.J.D., Giffin, A.L., Simmons, B.A., Hagger, V., Phua, C. and Hoegh-Guldberg, O. (2021). Identifying management opportunities to combat climate, land and marine threats across less climate exposed coral reefs. *Conservation Biology* (IF = 6.5, citations 2)

20. Roberson, L., Beyer, H.L., O'Hara, C., Watson, J.E.M., Dunn, Daniel, Klein, C.J., Frazier, M., **Kuempel, C.D.**, Williams, B., Grantham, H., Montgomery, J., Kark, S., Runting, R.K. (2021). Multinational coordination required for conservation of at least 90% of marine species. *Global Change Biology* (IF = 12.3, citations 16)

19. **Kuempel, C.D.**, Chauvenet, A.L.M., Symes, W.S., Possingham, H.P.P. (2021). Predicted protected area downsizing decreases habitat representation across terrestrial ecoregions in the tropics and subtropics. *Conservation Science and Practice* (IF = 3.3, citations 3)

18. Suarez-Castro, A.F., Beyer, H.L., **Kuempel, C.D.**, Linke, S., Borelli, P., Hoegh-Guldberg, O. (2021). Global forest restoration opportunities to foster coral reef conservation. *Global Change Biology* (IF = 12.3, citations 16)

17. Cottrell, R.S., Metian, M., Froehlich, H.E., Blanchard, J.L., Jacobsen, N.S., McIntyre, P.B., Nash, K.L., Williams, D.R., Bouwman, L., Gephart, J.A., **Kuempel, C.D.**, Moran, D.D., Troell, M., Halpern, B.S. (2021) Time to rethink trophic levels in aquaculture policy. *Reviews in Aquaculture* (IF = 11.1, citations
36) *Sena De Silva Award for novelty and impact*

16. **Kuempel, C.D.**, Froehlich, H.E., Halpern, B.S. (2021) An informed though experiment exploring the potential for a paradigm shift in aquatic food production. *Ocean and Coastal Management* (IF = 4.3, citations 6)

15. Adams, V.M., Dimitrova, N., Possingham, H.P., Allan, J.R., **Kuempel, C.D.**, Peterson, N., Kaiye, A. Keako, M., Tulloch, V.J.D. (2021) Scheduling incremental actions to build a comprehensive national protected area netowrk for Papua New Guinea. *Conservation Science and Practice* (IF = 3.3, citations
7)

14. Chauvenet, A.L.M., Watson, J.E.M., Adams, V.M., Di Marco, M., Venter, O., Davis, K.J., Mapping, B., Klein, C.J., **Kuempel, C.D.**, Possingham, H.P. (2020). To achieve big wins for conservation, prioritise protection of ecoregions closest to meeting targets. *One Earth* (IF = 16.7, citations 17)

13. **Kuempel, C.D.**, Frazier, M., Nash, K.L., Sand Jacobsen, N., Williams, D.R., Blanchard, J.L., Cottrell, R.S., McIntyre, P.B., Moran, D., Bouwman, L., Froehlich, H.E., Gephart, J.A., Metian, M., Tobben, J., Halpern, B.S. (2020). Integrating life-cycle and impact assessments to map food's cumulative environmental footprint. *One Earth* (IF = 16.7, citations 13)

12. Daigle, R., Metaxas, A., Balbar, A., McGowan, J., Treml, E., **Kuempel, C.D.**, Possingham, H., Beger, M. (2020). Operationalizing ecological connectivity to identify spatial conservation priorities with Marxan Connect. *Methods in Ecology and Evolution*, (IF = 9.2, citations 87) *Top cited paper 2022*

11. **Kuempel, C.D.**, Chauvenet, A.L.M., Possingham, H.P., Adams, V.M. (2020). Evidence-based guidelines for prioritizing investments to meet international conservation objectives. *One Earth* (IF = 16.7, citations 16)

10. Halpern, B.S., Blanchard, J.L., Bouwman, L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Sand Jacobson, N.,**Kuempel, C.D.**, McIntyre, P.B., Metian, M., Moran, D., Nash, K.L., Tobben, J., Williams, D.R. (2019). Opinion: Putting all food on the same table: Achieving sustainable food systems requires full accounting. *Proceedings of the National Academy of Sciences* (IF = 12, citations 74)
9. **Kuempel, C.D.**, Jones, K.J., Watson J.E.M., Possingham, H.P. (2019). Quantifying biases in placement of marine protected areas relative to abatable biodiversity threats. *Conservation Biology* (IF = 6.5, citations 39) *Top cited paper 2019*
8. Jantke, K., **Kuempel, C.D.**, McGowan, J.M., Chauvenet, A.L.M., Possingham, H.P. (2018). Metrics for evaluating representation target achievement in protected area networks. *Diversity and Distributions* (IF = 5, citations 24)
7. Jones, K.J., Klein, C.J., Halpern, B.S., Venter, O., Grantham, H., **Kuempel, C.D.**, Shumway, N., Friedlander, A.M., Possingham, H.P., Watson, J.E.M. (2018). The location and protection status of Earth's diminishing marine wilderness. *Current Biology* (IF = 11, citations 240)
6. **Kuempel, C.D.**, Adams, V.M., Possingham H.P., Bode M. (2018) Bigger or better: the relative benefits of protected area network expansion and enforcement for the conservation of an exploited species. *Conservation Letters* (IF = 8.3, citations 41)
5. Campbell, J.E., Altieri, A.H., Johnston, L., **Kuempel, C.D.**, Paperno, R., Paul, V.J., Duffy, J.E. (2018) Herbivore community determines the magnitude and mechanism of nutrient effects on subtropical and tropical seagrass. *Journal of Ecology* (IF = 6.3, citations 36)
4. Chauvenet, A.L.M., **Kuempel, C.D.**, McGowan J., Beger, M., Possingham, H.P. (2017). Methods for calculating Protection Equality for conservation planning. *PloS one* (IF = 3.8, citations 40)
3. **Kuempel, C.D.** and Altieri A. H. (2017). The emergent role of small-bodied herbivores in pre-empting phase shifts on degraded coral reefs. *Scientific Reports* (IF = 4.9, citations 58)
2. **Kuempel, C.D.**, Chauvenet, A.L.M., and Possingham H.P. (2016). Equitable representation of ecoregions is slowly improving despite strategic planning shortfalls. *Conservation Letters* (IF = 8.3, citations 41)
1. Nelson, H.R., **Kuempel, C.D.**, and Altieri, A.H. (2016). The resilience of reef invertebrate biodiversity to coral mortality. *Ecosphere* (IF = 3.4, citations 50)

Grants* and consulting†

DFAT Australia Awards Nature-based solutions short course in Papua New Guinea † (2025)

WWF modelling sediment run-off for Fiji's Great Sea Reef ($40k) † (2025)

Bush Heritage Bespoke R Training Course ($16.5k) † (2024)

Inspiring Queensland National Science Week Grant (Brisbane Soapbox Science) † ($4k) (2024)

WWF Green Climate Fund Fiji Feasibility Study ($55k) † (2023-2024)

SPREP Kiwa Initiative Micro-credentials ($135,335) † (2024)

Future Earth Australia Creating Opportunities Grant ($6k)* (2023)

Griffith Advancing Women's Research Success Grant ($4,601.73)* (2023)

Blue Economy CRC Marine Spatial Planning (responsible for $270k in post-doc salary)* (2023)

Global Environmental Facility Coral Reef Rescue project (US$7M)* (2022)

CBCS Small Grants Scheme ($4,500 across three successful proposals)* (2021)

ARC Laureate Postgraduate Scholarship ($119,376 over 4 years)	Nov 2014-2018
Seafood Footprint Displacement Top-up Scholarship (Dr. Carissa Klein, $5,000)	March-June 2018
50 Reefs Top-up Scholarship ($5,000)	Sept. 2017
University of Queensland Career Development Award ($5,000)	May 2017
Great Barrier Reef Marine Park Authority Science for Management Award ($1,000-declined)	May 2017
Australian Coral Reef Society Writing Retreat Scholarship ($500)	May 2016
Smithsonian Tropical Research Institute Short-Term Fellowship ($2,000)*	June-Nov. 2013
University of Iowa Old Gold Scholarship ($12,000)	Aug. 2007-May 2011
University of Iowa Athletic Scholarship (Rowing)	Jan. 2008-May 2011

Teaching Experience

3606ENV Conservation Biology - course convener, lecturer

> ~110 students (T1 2023 and 2024), SEC/SET >4 with feedback that I am "knowledgeable and excited about the content of the course and presents lecture material in a way that makes it easy to learn."

3804ENV Marine Ecosystem Modelling - course convener, lecturer

> ~30 students (T1 2023 and 2024), SEC/SET >4 with feedback that "I loved this course! The material I found very interesting and challenging. This course made me realise this is the avenue I want to go down after graduation"

3033ESC Environmental Data Analytics - lecturer

> 4 students (T2 2023), SEC/SET 5 (1 response) with feedback that "Caitie's teaching was great, she was patient and always willing to help. She would often push us to figure out things by giving hints to help guide you without just giving away the answer which I really liked. Definitely one of the best studio/tutorial teachers I've had during my degree."

ENVM2002/7002 Marine and Coastal Environmental Protection, University of Queensland,	August 2021
ENVM3220/7202 Conservation Planning and Management, University of Queensland,	March 2021
WWF Structured Decision Making Training (6 weeks), online,	April-May 2021
ENVM2002/7002 Marine and Coastal Environmental Protection, University of Queensland,	August 2020
CONS7021 Masters conservation decision-making course, University of Queensland	June 2017, 2018
Moreton Bay Research Station Science Camp, University of Queensland	Oct. 2017-June 2018
BIOL3000 Third year conservation course, University of Queensland	Aug. 2017
Spatial planning and decision-science workshop, Bogota, Colombia	July 2017
Standford University Bing Overseas Program, University of Queensland	Nov. 2017
R for beginners, WWF Indonesia Jakarta office	Oct. 2016
Spatial planning for marine protected area design, Bali, Indonesia	Oct. 2016
MARS2005 Australia's Marine Environments second year course, University of Queensland	May 2016

Supervision (* denotes primary supervisor)

Ainsley Leaning *, Masters (2025 - present)

Ecosystem characteristication for marine spatial planning using hydroacoustics

Jade Arnold*, Masters (2024 - first class)

The conservation potential of Australia's marine protected area network

Simon Metcher*, Honours Student (2024- first class)

Commencing PhD under my supervision in April 2025 Assessing bivalve farming potential to mitigate nutrient pollution in the Great Barrier Reef

Siaan Charles*, Advanced Studies (2024)

Methods for First nations engagement in Marine Spatial Planning

Dr Jackson Stockbridge*, Post-doctoral fellow (2023-present)

Developing a framework for assessing cumulative impacts of the blue economy in Australia

Marina Christofidis*, PhD Student (2023-present)

Climate risk to aquaculture production in Australia (international scholarship recipient)

Alana McPherson, Honours Student (2023-2024)

Conservation benefits of Marine Protected Areas over time in Australia

Sophie Petrie*, Capstone project, (2023)

The Potential Impacts of Aquaculture on Threatened Species in Queensland

Emma Arnett, Master's Student (2020-2021)

Quantifying like-for-like global seafood trade Now employed at DETSI

Ella Sinclair, Volunteer (2020)

Primary supervisor for conducting literature review of monitoring and evaluation indicators for WWF Coral Reef Rescue Initiative Now employed as an environmental consultant

Awards

Vice Chancellor's Research Excellence Award – Early Career Researcher nominee (1 of 4 nominees) (2023)

Foundation for the WA Museum Fame Lab Semi-finalist: To lower your environmental footprint consider what your dinner ate (2023)

University of Queensland Faculty of Science Awards Rising Star - Academic category Finalist (Dec. 2020)

APEC Science Prize for Innovation, Research and Education Award - shortlisted for Australian nominee (2020)

University of Queensland Dean's Award for Outstanding HDR Theses (2018)

Taylor and Francis University of Queensland Library Excellence Award in Research ($1,000) May 2017

Other Publications and Works

Stockbridge, J., **Kuempel, C.D.** (2024) Identifying synergies and trade-offs between sectors in the Blue Economy Zone. Blue Economy Cooperative Research Centre Milestone Report.

Stockbridge, J., **Kuempel, C.D.** (2024) Evaluation of potential for interaction and/or conflict among ocean users and the environment in offshore waters in Australia. Blue Economy Cooperative Research Centre Milestone Report.

Klein, C, **Kuempel, C.D.**, Roberson, L., Cottrell, R. (2023) Submission to Measures to prevent the importation of illegal, unreported and unregulated seafood discussion paper. Australian Government Department of Agriculture, Fisheries and Forestry

Wenger, A. . . ..**Kuempel, C.D.** et al. (2023). A guide for Integrated Conservation and Sanitation Programs and Approaches. Wildlife Conservation Society.

Kuempel, C.D. (2023) Sedimentation sifted out of global pollution priorities. *Science* (IF = 54.5)

Kuempel, C.D. (2023) Farmed salmon or chicken? Environmental footprint research can guide eco-conscious consumers. The Conversation

Almaraz, M. and **Kuempel, C.D.** (2022) The impact of excessive protein consumption on human wastewater nitrogen loading of US waters. Global Water Forum

Kuempel, C.D. and Alyssa Giffin (2021) Resilient reefs, resilient communities: how can we get there? Catchment to Coast

Jones, K., Friedlander, A. Halpern, B., **Kuempel, C.D.**, Klein, C. Grantham, H., Possingham, H., Watson, J., Shumway, N., Venter, O. (2018) New map shows that only 13% of the oceans are still truly wild. The Conversation

Kuempel, C.D. Southern Coral Sea (2017) Reef Life Survey blog

Kuempel, C.D., McGowan J., Estradivari, Handaynai C., Daniel D., Possingham H., Ahmadia, G. (2017) Conservation priority gaps in the Sunda Banda Seascape, Indonesia

Kuempel, C.D., Di Marco M., Watson J., Possingham H. (2017). Achieving the targets of global conventions. *Decision Point* http://decision-point.com.au/article/achieving-the-targets-of-global-conventions/

Select Media and Outreach

As an example of reaching a broad non-scientific audience, my written work for The Conversation blog (theconversation.com) has had >200,000 readers.

Queensland prawn areas at risk of inundation by rising seas this century The Conversation

Which food is better for the planet? The Washington Post

Food for thought: the carbon footprint of salmon and chicken farming mostly stems from feed. The Guardian

To lower food emissions consider what your dinner ate. Popular Science

Farmed chicken and salmon have a global footprint. Nature

Coming out or not, earthquake aid, and what's the most eco-friendly meat? Triple J Hack

Female scientists encouraging children to embrace science Courier Mail

Presentations

* *oral presentation,* † invited presentation

Kuempel C† (2024) Frameworks to assess cumulative impacts of Australia's Blue Economy *International Marine Conservation Congress, Cape Town, South Africa*

Kuempel C† (2024) The need and feasibility of the Fiji Coral Reef Resilience project *International Marine Conservation Congress, Cape Town, South Africa*

Kuempel C† (2024) Frameworks to assess cumulative impacts of Australia's Blue Economy *Australian Marine Science Association*

Kuempel C† (2023) Frameworks to assess cumulative impacts of Australia's Blue Economy *Blue Economy CRC Participants Workshop, Kingscliff, Australia*

Kuempel C† (2023) Putting Australian Aquaculture on the Map *Australian Marine Sciences Association, Gold Coast, Australia*

Kuempel C† (2022) Mapping a path for conservation in a developing world. *Griffith University, Brisbane, Australia*

Kuempel C† (2022) Conservaiton nature in a developing world. *Linn-Mar High School, Marion, IA, USA*

Kuempel C (2021) Informing management across less-climate exposed coral reefs. *International Congress on Conservation Biology*, on-line

Kuempel C† (2021) Coral Reef Rescue Initiative: A climate-centric approach for managing coral reefs for nature and people. *Coral Reef Rehabilitation and Management Program - Coral Triangle Initiative*, on-line

Kuempel C† (2020) The basics of protected area design and evaluation. *University of Queensland Twilight Talk Series*, on-line

Kuempel C† (Nov. 2020) Getting Started with GitHub in R. *Pete Mumby lab group meeting*, UQ

Kuempel C† (Oct. 2020) Workflows and best practices for collaborative coding. *RLadies Brisbane*, online

Kuempel C† (Sept. 2020) Getting Started with GitHub in R. *RLadies Brisbane*, online

Kuempel C† (2020) Coral Reef Rescue Initiative: A climate-centric approach for management coral reefs for nature and people. *Interanational Marine Conservation Congress*, on-line

Kuempel C† (2019) Conserving nature in a developing world. *NCEAS Roundtable*, Santa Barbara, CA, USA

Kuempel C, Adams V, Possingham H, Bode M (2018) Bigger or better: the relative benefits of MPA expansion and enforcement. *Society for Conservation Biology Oceania Conference*, Wellington, New Zealand

Kuempel C, Jones K, Watson J.E., Possingham H (2018) Global marine protected area establishment largely avoids abatable threats to biodiversity. *International Marine Conservation Congress*, Kuching, Malaysia

Kuempel C† (2018) Evaluating trade-offs between quantitative and qualitative protected area objectives. *Centre for Biodiversity and Conservation Science Seminar*, University of Queensland

Kuempel C, Adams V, Possingham H, Bode M (2017) Bigger or better: the relative benefits of MPA expansion and enforcement. *4th International Marine Protected Areas Congress*. La Serena, Chile

Kuempel C, Jones K, Watson J.E., Possingham H (2017) Are marine protected areas tackling threats? *International Congress on Conservation Biology*, Cartagena, Colombia

Kuempel C, Chauvenet A, Possingham H (2016) Does rapid protected area expansion impact the ecological representation of reserve netwowrks? *Society for Conservation 4th Oceania Congress*, Brisbane, Australia

Kuempel C, Bode M, Adams V, Possingham H (2016) Balancing area and effectiveness: MPA expansion vs. management enforcement in meeting global conservation targets.*International Coral Reef Symposium*, Honolulu, HI

Kuempel C† (2016) Balancing area and effectiveness: Investigating trade-offs in the Convention on Biological Diversity Aichi Target 11. *Environmental Science Seminar*, University of Iowa

Kuempel C, Chauvenet A, Possingham H (2016) Does rapid protected area expansion compromise ecological representation? *Student Conference on Conservation Science*, University of Cambridge, UK

Kuempel C and Altieri A (2014) The role of consumers in mediating coral-algal phase shifts. *43rd Annual Benthic Ecology Meeting*, Jacksonville, FL, USA

Service Roles

Academic

International Congress on Conservation Biology 2025 *Steering Committee Member*

School of Environment and Science *Early Career Representative* (2024-present)

Australian Rivers Institute *Early Career Researcher Representative* (2023)

Griffith Women in Leadership 2023 Cohort

Centre for Biodiversity and Conservation Science Management Committee *Early Career Researcher Education Lead* (2021-Present)

UNEP MPA Toolbox and Learning Platform *Academic Associate, Content editor/coordinator/contributor*

NCEAS Diversity Team representative (2018-2019)

Volunteer Researcher: University of Queensland and WWF Indonesia Sept.-Oct. 2016

Reviewer for international publications

> Nature Sustainability, the Journal of Environmental Management, Biological Conservation, Marine Biology, PLoS One, Global Ecology and Conservation, Coral Reefs, Conservation Letters, Conservation Biology, Diversity and Distributions, Marine Policy, Conservation Science and Practice and PeerJ.

Queensland Catch a Rising Star Program, Queensland National Science Week (2016)

PhD seminar series Organiser (2014-2015), The University of Queensland, St. Lucia, QLD, Australia

Community

Soapbox Science Brisbane co-organiser (2024-present) - community engagement to promote women in STEM during National Science Week

Certified Software Carpentries Instructor (2020-present) - specialising in introductory to advanced R programming skills and version control (e.g., GitHub).

RLadies Brisbane Oraganiser - promoting women in programming (2020-2024)

Marxan Spatial Planning Instructor (2017-current) - successful workshops in Australia, Indonesia, Colombia, and online

Professional Membership

Australian Marine Sciences Association

Reef Life Survey Volunteer (Coral Sea expedition 2018)

Australian Coral Reef Society

Society for Conservation Biology > *Board Member/Treasurer* (2015-2017)

International Society for Reef Studies

World Commission on Protected Areas Young Professionals Group

Working Groups

Organiser/Participant Australia's sustainable seafood future, UQ Heron Island Research Station, April 2022

Organiser/Participant Centre for Biodiversity and Conservation Science HDR/ECR Writing workshop, Moreton Bay Research Station, Dec. 2021

Participant Science for Nature and People Partnership (SNAPP) Coastal Health working group, NCEAS (on-line), Nov. 2021

Organiser/Participant CBCS Integrated Watershed Management Workshop, University of Queensland, Oct. 2021

Panel Participant Working with "big data" in the biological sciences, University of Queensland, July 2021

Organiser/Participant Coral Reef Rescue Initiative Near Real-Time Monitoring Workshop, WWF online, Sept. 2021

Participant Strengthening coastal communities resilience by supporting community enterprises, WWF online, Oct. 2020

Participant What's the catch? Social and environmental sustainability of seafood, Heron Island, University of Queensland Jan 2021

Post-doctoral Fellow Comparing the environmental impacts of global food systems, NCEAS, UC Santa Barbara Sept. 2018-2020

Participant Sustainable seafood and marine conservation, Heron Island, University of Queensland June 2019

Participant Incorporating connectivity into spatial conservation planning,The Nature Conservancy, Washington D.C. 2017

Participant/Coordinator Land-sea conservation planning in Papua New Guinea, Port Moresby, Papua New Guinea 2017

Participant/Coordinator Marine spatial planning in the Sunda Banda Seascape, Bali, Indonesia 2016

Organizer/Participant Global Marine Conservation. Lady Elliot Island, Australia 2016

Skills

Programming: = R (*advanced*), Github (*advanced*), Matlab (*basic*) *Software:* = Marxan (*advanced*), Arc/QGIS, Adobe Illustrator, Microsoft Office